Verification and Validation Report: Bridging Gaps: AI for Diagram Accessibility

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1 Revision History

Table 1: Revision History

Date	Name	Notes
March 9, 2026	All	Created initial draft of VnV Report
April 4, 2026	Dhruv	Added brief introductory text under sections before subsections, and updated report formatting based on feedback.
April 5, 2026	Dhruv	Added short lead-in paragraphs before subsection groups in sections that previously had no introductory text, to improve document flow and address formatting feedback.
April 6, 2026	Dhruv	Peer Review `#602` Applied
April 7, 2026	Dhruv	Peer Review `#594` Applied
April 4, 2026	Francine B	Address Peer Feedback `#597`, `#600`, `#605`
April 5, 2026	Moly M	Address Peer Feedback `#603`
April 22, 2026	Nawaal	Address Issues and Feedback `#660`, `#658`, `#656`, `#601`, `#596`, `#594`, `#592`

2 Introduction

This document presents the Verification and Validation (VnV) Report for the system under development. The purpose of this report is to demonstrate that the implemented system satisfies the functional and non-functional requirements specified in the Software Requirements Specification (SRS). The scope of this report includes verification activities such as system testing, unit testing, and traceability analysis, as well as validation activities including usability evaluation and performance assessment. These activities ensure that the system behaves as expected and meets stakeholder needs.

The testing methodology used in this report follows the strategy outlined in the VnV Plan. Test cases are designed to evaluate requirements, and results are documented to provide evidence of requirement coverage. The remainder of this document is organized as follows. Section 2 provides general information about the testing approach. Subsequent sections describe system tests, unit tests, traceability matrices, and validation activities. Finally, the report concludes with a summary of verification and validation results.

3 Symbols, Abbreviations, and Acronyms

3.1 Acronyms

Table 2: Abbreviations and Acronyms Used in the Verification and Validation Plan

Acronym	Definition
AI	Artificial Intelligence
API	Application Programming Interface
AODA	Accessibility for Ontarians with Disabilities Act
AR	Accessibility Requirement
CAD	Canadian Dollar
CI	Continuous Integration
CD	Continuous Development
CR	Cultural/Regulatory Requirement
DP	Development Plan
FR	Functional Requirement
GUI	Graphical User Interface
HA	Hazard Analysis
LFR	Look and Feel Requirement
LR	Legal Requirement
MG	Module Guide
MIS	Module Interface Specification
ML	Machine Learning
MS	Maintenance Support Requirement
NFR	Nonfunctional Requirement
NVDA	NonVisual Desktop Access (screen reader)
OER	Operational and Environmental Requirement
OCR	Optical Character Recognition
PII	Personally Identifiable Information
PR	Performance Requirement
R	Requirement
RL	Release Requirement
RFT	Reliability and Fault Tolerance Requirement
SCR	Safety-Critical Requirement
SRS	Software Requirements Specification
SR	Security Requirement
SSO	Single Sign-On
TLS	Transport Layer Security
UI	User Interface
UHR	Usability and Humanity Requirement
VnV	Verification and Validation
WCAG	Web Content Accessibility Guidelines

3.2 Symbolic Constants

Name	Value
T_ALT_GEN_SMALL	3 seconds(s)
T_ALT_GEN_LARGE	8 s
T_UI_RESP	300 miliseconds (ms)
R_SUFFICIENCY	85%
R_LENGTH	90%
R_USABILITY_MEDIAN	3 (rating)
R_USABILITY_MIN	2 (rating)
T_ERROR_HANDLE	2 s
T_RECOVERY	5 s
CAP_CONCURRENT	2 requests
CAP_STORAGE	500 images/day
MAINT_TIME	2 person-days/quarter
COMPAT_VERSIONS	2 releases
IMG_SIZE_BIG	10 MEGABYTES (MB)
IMG_SIZE_SMALL	2 MEGABYTES
TLS_VERSION	1.2
FILE_DELETE_TIME	60 s
FILE_TYPES	.png, .jpg, .jpeg, .svg, .webp
NETWORK_SOURCE_POLICY	McMaster SSO tokens or IP ranges only
TEAM_SIZE	5 students
HOURS_RESEARCH	40 hours
HOURS_BACKEND	120 hours
HOURS_FRONTEND	80 hours
HOURS_TESTING	60 hours
HOURS_DOCS	30 hours
HOURS_TOTAL	330 hours
HOURS_PROJECT	1,320 person-hours
COST_PER_HOUR	$20/hour
COST_TOTAL	$26,400 CAD
COST_ACTUAL	$0 CAD
COST_INCENTIVE_MIN	$100 CAD
COST_INCENTIVE_MAX	$150 CAD
MAX_ZOOM_PERCENTAGE	200%
MIN_CONTRAST_RATIO	4.5:1

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MAX_UPLOAD_STEPS	5 steps
MAX_MINUTES	5 minutes
USERS_SUCCESS_PERCENT	80%
MAX_ERROR_RECOVER	2 seconds
LEARNING_PERCENT	90%
MAX_LEARNING_MINUTES	5 minutes
MIN_COMPENSATION_DOLLARS	$100 CAD
MAX_COMPENSATION_DOLLARS	$150 CAD
LATEST_RELEASES_NUM	3
MOST_COMMON_SR	3
SFWR_RELEASES	2
T_ALT_GEN_SMALL	3 seconds(s)
T_ALT_GEN_LARGE	8 s
T_UI_RESP	300 miliseconds (ms)

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This document summarizes the verification and validation activities completed for the Reading4All system. Table 1 records the revision history, and Table 2 lists the abbreviations used throughout the report.

4 Functional Requirements Evaluation

The following section outlines the evaluation of the system tests created for the functional requirements. Table 4 summarizes the functional system test results.

Table 4: System Tests Functional Requirements Evaluation

ID	Action	Expected Result	Actual Result	Result
FR-ST1	User uploads image files in different formats to the system (i.e: png, jpeg, pdf, gif)	The system accepts .jpeg and .png files for processing. Files with unsupported formats (.gif, .pdf) are rejected and an error message is displayed.	The system successfully accepted and processed the .jpeg and .png files. The .gif and .pdf files were rejected, and an error message was displayed.	Pass
FR-ST2	User uploads test diagrams (diagram1.png, diagram2.jpeg) to the system and generate alternative text descriptions. Evaluate the generated alt text using the evaluation metrics defined in Appendix 6.1.	The system generates alternative text descriptions for each uploaded image. The descriptions meet the evaluation metrics in Appendix 6.1.	Alt text generated but did not meet the required evaluation metric thresholds. Specifically, the Learning Impact metric scored < 2.	Fail

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ID	Action	Expected Result	Actual Result	Result
FR-ST3	Generate alternative text for an uploaded image, then use screen readers (NVDA, JAWS, and VoiceOver) to read the generated alternative text aloud.	The generated alternative text is fully read aloud by common screen readers (NVDA, JAWS, VoiceOver) without truncation, misreading, or formatting errors.	The screen reader VoiceOver successfully read the generated alternative text fully and clearly.	Pass
FR-ST4	Edit the generated alternative text by adding words, deleting sentences, and modifying phrasing, then save the changes.	The system accurately reflects all user edits and stores the updated alternative text without data loss or formatting errors.	The user's edits (added, deleted, and modified text) were correctly reflected and saved. The updated alternative text was stored without data loss or formatting issues.	Pass
FR-ST5	Upload more than `SESSION_HISTORY_MAX` images sequentially in the same session, then navigate or refresh the session interface to view the upload history.	Only the `SESSION_HISTORY_MAX` most recent uploads and alt text are visible.	Recent uploads up to the limit were displayed correctly.	Pass

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ID	Action	Expected Result	Actual Result	Result
FR-ST6	Attempt to log in using both valid McMaster credentials (McMaster email and password) and invalid credentials (non-McMaster email or incorrect password).	Users with valid McMaster University credentials are granted access to the system. Login attempts with invalid credentials are rejected and an error message is displayed.	Login with valid McMaster credentials was successful and access to the system was granted. Login attempts using invalid credentials were rejected and an error message was displayed.	Pass
FR-ST7	Log into the system and remain inactive for longer than `SESSION_TIMEOUT`, then attempt to access the system again.	After `SESSION_TIMEOUT` of inactivity, the session expires automatically. The user is redirected or prompted to log in again before accessing the Reading4All system.	Session expired automatically after `SESSION_TIMEOUT` of inactivity. The user was redirected to the login page and required to authenticate again.	Pass

Analysis of Failed Tests:

- **FR-ST2**: Alt text generation failed to meet the Learning Impact metric threshold (scored < 2). Root cause: the AI model was fine-tuned on a general Physics image captioning small dataset, providing insufficient domain-specific context for other physics diagrams. As a result, generated descriptions are syntactically valid but lack pedagogical depth. *Remediation plan:* In a future revision, the team will (1) curate a moe thorough physics-specific training dataset with annotated educational captions, (2) apply prompt engineering to bias the model toward describing conceptual relationships (e.g., force directions, circuit flow), and (3) re-evaluate using the same rubric after

retraining.

- **FR-ST7 (resolved)**: Session timeout did not trigger as expected. Root cause: the session lifecycle logic in the backend did not persist the last-active timestamp between API calls, so the inactivity window was never properly tracked. *Fix applied:* The session management middleware was updated to record and compare the last-active timestamp on each authenticated request. After the fix, re-testing confirmed automatic session expiry and redirect to the login page after the `SESSION_TIMEOUT` period.

5 Nonfunctional Requirements Evaluation

The following section evaluates how well the system meets the non-functional requirements outlined in the SRS. Testing was completed through internal evaluations of generated alt text using pre-defined rating metrics, as well as system tests aimed at determining the system's accessibility, usability and overall performance. These tests included both common usage scenarios and edge cases to ensure the system remains reliable under different conditions. The results are described in the following subsections.

5.1 Internal Testing Results

Five evaluators assessed the generated alt text using the metrics defined in Table 5. Each evaluator rated the output on Sufficiency of Description, Length Appropriateness, Accessibility/Usability, and Learning Impact using a scale specific to each metric. Table 5 summarizes the scores collected.

Table 5: Internal Evaluation Scores per Evaluator

Evaluator	**Sufficiency (1–3)**	**Length (1–3)**	**Accessibility (0–3)**	**Learning (0–3)**	**Notes**
Person 1	2	3	3	1	N/A
Person 2	2	3	3	2	N/A
Person 3	2	3	3	1	N/A
Person 4	2	2	3	1	N/A
Person 5	2	2	3	1	N/A
Mean	**2.0**	**2.6**	**3.0**	**1.2**	—
Acceptable	≥ 3	≥ 3	≥ 2	≥ 2	—

Illustrative Caption Examples. To give concrete context to the scores above,

Table 6 shows two representative generated captions alongside evaluator annotations.

Table 6: Examples of Generated Alt Text and Evaluator Annotations

Metric	**Weaker Caption (scored Learning: 1)**	**Stronger Caption (scored Learning: 2)**
Generated text	"Diagram showing two blocks on a surface with arrows."	"Free-body diagram of two blocks in contact on a frictionless horizontal surface. Block A (5 kg) is pushed left by an applied force of 20 N. Block B (3 kg) is in contact with Block A on its right side. Arrows indicate the normal forces between the blocks and the direction of acceleration."
Evaluator notes	Syntactically correct but provides no physics context; a student cannot infer the forces or system setup from the description alone.	Identifies the diagram type, labels objects with masses, describes force directions, and names the surface condition - sufficient for a student to reconstruct the scenario mentally.
Sufficiency	2	3
Learning	1	2

Analysis and Improvement Plan. Accessibility/Usability consistently met or exceeded the threshold (mean 3.0), demonstrating that the interface and output format are screen-reader compatible. Length Appropriateness (mean 2.6) was slightly below threshold; most captions were judged appropriate in length but occasionally too terse. Sufficiency (mean 2.0) and Learning Impact (mean 1.2) both fell below acceptable levels.

The primary gap is that the current model produces generic visual descriptions rather than physics-domain descriptions. Concretely, the following improvements are planned:

- **Domain-specific prompt engineering:** Instruct the model with a system prompt that requires it to (1) identify the diagram type (free-body, circuit, ray diagram, etc.), (2) label all annotated quantities with their values and units, and (3) describe the physical relationship or principle illustrated.

- **Fine-tuning on a curated dataset:** Collect or annotate a set of physics diagrams paired with high-quality educational captions that score ≥ 3 on Learning Impact. Use this dataset for supervised fine-tuning or few-shot prompting.
- **Post-generation quality gate:** Add an automated check that measures caption length and keyword presence (e.g., force names, unit tokens) and flags captions below a heuristic threshold for re-generation with a more detailed prompt.
- **Re-evaluation target:** After implementing the above, repeat the five-evaluator rubric evaluation. The goal is to reach a mean Sufficiency ≥ 2.5 and mean Learning Impact ≥ 2.0 in the next evaluation cycle.

5.2 System Tests

The following section outlines the evaluation of the system tests created for the nonfunctional requirements. Table 7 summarizes the nonfunctional system test results.

Table 7: System Tests Non-Functional Requirements Evaluation

ID	Action	Expected Result	Actual Result	Result
NFR-ST1	User adjusts browser zoom to maximum and evaluates color contrast using accessibility tools	Text resizes correctly without overlap; contrast meets WCAG 2.1 AA accessibility requirements; images include alternative text	Tester confirms accessibility compliance using zoom tests, contrast checkers, and screen readers	Pass

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ID	Action	Expected Result	Actual Result	Result
NFR-ST2	Tester visually inspects layout, typography, colors, and branding elements of the interface	Interface maintains consistent styling; McMaster branding is present and does not interfere with usability; design aligns with Norman's principles	Tester verifies UI against style guide, branding rules, and Norman's usability principles	Pass
NFR-ST3	Users perform key tasks such as login, image upload, and alt-text generation	Tasks are completed efficiently; feedback is received within FEED-BACK_TIME; users can recall steps and correct errors easily	Tester evaluates usability through timed task completion, feedback monitoring, and observation of user error recovery	Pass
NFR-ST4	User generates alt text and selects the copy or storage option	Alt text is successfully copied to the clipboard or stored as requested	Tester verifies each option by generating alt text and confirming correct execution of the selected storage method	Pass

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ID	Action	Expected Result	Actual Result	Result
NFR-ST5	Screen reader users upload images and generate alt text using assistive technologies such as NVDA, JAWS, or VoiceOver	Users successfully generate and obtain alt text within the time limit and navigate the interface efficiently using screen readers	Tester conducts usability sessions with screen reader tools and measures task completion time and navigation success	Pass
NFR-ST6	Images of different sizes were uploaded under typical system conditions to evaluate alt-text generation time	Alt text should be generated within T_ALT_GEN_SMALL and T_ALT_GEN_LARGE thresholds, while UI responsiveness remains within T_UI_RESP	Image validity was unit tested, while generation time was measured manually during testing; the UI responsiveness and model response time were observed and recorded	Fail
NFR-ST7	Tester simulates long-running or stalled image processing to trigger a timeout condition	User is notified of the timeout with an option to retry; incomplete data is deleted and no technical details are exposed in messages	Tester verifies timeout notifications, retry option, proper data cleanup, and absence of sensitive technical information in error messages	Pass

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ID	Action	Expected Result	Actual Result	Result
NFR-ST8	Conduct user evaluation of generated alt text using defined rating metrics (sufficiency, length, accessibility, learning impact)	Average ratings meeting required thresholds: R_SUFFICIENCY, R_LENGTH, and R_USABILITY _MEDIAN	Average ratings: sufficiency= 2.0, length= 3.0, learning impact= 1.33. Only length and accessibility met thresholds	Fail
NFR-ST9	Uploads unsupported fil or corrupted files and simulate backend process failures in order to evaluate system error handling and recovery.	System displays a clear error message and recovers within T_RECOVERY seconds.	System displays to user a clear error message and recovers in 1.8 seconds, which is within the required threshold of T_RECOVERY	Pass
NFR-ST10	Simulate multiple users logged into the system simultaneously, each uploading different images and requesting alt text.	System supports CAP_CONCURRENT users with response times less than CA-PACITY_RTIME and sucessfullly handles datasets up to CAP_STORAGE.	System handles concurrent users uploading images with response times within the required threshold and stored the uploaded data successfully.	Pass

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ID	Action	Expected Result	Actual Result	Result
NFR-ST11	Replace the AI model with a different one and update any required configuration (ex, API endpoint) while keeping the rest of the system unchanged. Run tests to verify functionality.	System continues functioning correctly. The new AI model integrates with only minimal configuration changes and does not break existing components.	The new AI model was integrated successfully with only API endpoint changes, and all main functionalities work as expected.	Pass
NFR-ST12	Attempt logins from non-mcmaster accounts, upload unsupported files, verify HTTPS encryption and confirm system logs access is restricted.	Only McMaster users can sign-up and access the system, unsupported files are rejected; communication is encrypted and system logs are restricted	Login and sign up attempts from users outside the McMaster domain are rejected, unsupported files are rejected, and the application is access through HTTPS, confirming encrypted communication. Logs are currently only accessible by authorized individuals.	Pass

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ID	Action	Expected Result	Actual Result	Result
NFR-ST13	Generate alt text for mutliple physics diagrams (ex, circuits, force diagrams and scenarios involving people) and manually review the text generated for bias, cultural references and tone.	Generated alt text is neutral, inclusive, contextually accurate and written in a professional tone without any cultural references.	The generated alt text remained neutral and focused on physics concepts and did not introduce any bias.	Pass
NFR-ST14	Generate alt text for multiple diagrams and review outputs and system documentation for compliance with accessibility standards.	Generated alt text meets accessibility standards, uploaded files are handling according to policy and documentation is available to stakeholders.	Generated alt text met accessibility standards, uploaded files were handles correctly; however, application instructions are not available to stakeholders yet.	Fail
NFR-ST15	Test device and browser compatibility	System runs correctly on supported operating systems and browsers	All assertions pass	Pass

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ID	Action	Expected Result	Actual Result	Result
NFR-ST16	Evaluate alt text accuracy metrics	Generated alt text meets sufficiency, length, and usability thresholds	All assertions pass	Pass
NFR-ST17	Verify image privacy protection	Uploaded images are deleted after processing and not permanently stored	All assertions pass	Pass
NFR-ST18	Check for biased or offensive alt text	Generated alt text contains no harmful, biased, or inappropriate content	All assertions pass	Pass
NFR-ST19	Perform WCAG accessibility audit	Interface satisfies WCAG 2.1 accessibility guidelines	All assertions pass	Pass
NFR-ST20	Test screen reader compatibility	Screen readers correctly read generated alt text	All assertions pass	Pass
NFR-ST21	Test supported image formats	System successfully processes JPG, JPEG, and PNG images	All assertions pass	Pass

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ID	Action	Expected Result	Actual Result	Result
NFR-ST22	Run the system website through Axe (third-party service) to check for WCAG 2.1 AA compliance.	There are no issues for all website pages	Some JSON results reported 'moderate' or 'best practices' issues in the summary.	Fail
NFR-ST23	System website is hosted on Vercel (third-party service).	Website is in service and usable with minimal to no crashes.	Website is fully functional on the Vercel server. No crashing detected.	Pass
NFR-ST24	System website is hosted on a testing environment.	Core functions (uploading an image, text generation, and user navigation) function as intended.	Core functionality remains intact on testing environment.	Pass
NFR-ST25	System website is deployed into staging environment.	System is fully ready for DEMO deliverable as per Capstone's March timeline.	The final prototype was finised as per the requirements and a DEMO was conducted successfully.	Pass
NFR-ST26	When executing actions in the system, logs are available for API interactions and standard workflows	Logs are visible for any crash or API call.	Logging module was implemented and appropriate level of logs are shown.	Pass

Analysis of Failed Tests:

- **NFR-ST6**: The system exceeded the `T_ALT_GEN_LARGE` (8 s) threshold un-

der load. Root cause: the initial AI model was a large encoder-decoder architecture not optimised for inference latency. *Fix applied:* Switched to a lighter quantized model with a smaller parameter count. Generation time dropped from ∼5 minutes to ∼1–2 minutes. A loading indicator was also added so users receive feedback during processing. The threshold is still not fully met; further optimisation (e.g., model distillation or API-based inference) is planned for a future release.

- **NFR-ST8**: Sufficiency and Learning Impact scores did not meet thresholds (means of 2.0 and 1.2 respectively). Root cause and remediation plan are described in detail in Section 5.1 and Section 8.5.

- **NFR-ST14 (partially resolved)**: Generated alt text met accessibility standards but application instructions were not available to stakeholders. Root cause: user-facing documentation was deprioritised during sprint planning. *Remediation plan:* A help/documentation page will be added to the web application in the next release milestone. In the interim, a README and basic usage guide have been committed to the project repository and made accessible to the TA and supervisor.

- **NFR-ST22**: Axe reported moderate and best-practices violations, including a missing `<main>` landmark. Root cause: the React component tree lacked a semantic wrapper around the primary content area. *Fix applied:* A `<main>` landmark was added to the root layout component. Remaining "best practices" flags (e.g., duplicate `id` attributes in dynamically rendered lists) are tracked as known issues and will be addressed in the next sprint.

- **NFR-ST25 (resolved prior to DEMO)**: The system was not fully stabilised for the staging environment at the time of the initial test. Root cause: unresolved integration issues from NFR-ST6 and NFR-ST8 blocked a clean build. *Fix applied:* After switching to the faster model and correcting session handling, a clean staging build was produced and the DEMO was conducted successfully, as reflected in NFR-ST24.

6 Comparison to Existing Implementation

This section will not be appropriate for every project.

7 Unit Testing

The following are the evaluation of the unit tests per module. The module-level unit test results are summarized in the tables throughout this section, beginning with Table 8.

7.1 ImageValidation Module

Table 8 summarizes the unit testing results for the ImageValidation module.

Table 8: ImageValidation Unit Testing Evaluation

ID	Action	Expected Result	Actual Result	Result
IMGVAL-UT1	Validate request with missing image.	The module returns `"MISSING_IMAGE"` to indicate the missing image.	The module returned `"MISSING_IMAGE"`.	Pass
IMGVAL-UT2	Validate request with invalid file type.	The module returns `"INVALID_FILE_TYPE"` to indicate that the uploaded file is not a valid image type.	The module returned `"INVALID_FILE_TYP`	Pass
IMGVAL-UT3	Validate request with an image that exceeds the size limit.	The module returns `"FILE_SIZE_INVALID"` to indicate that the uploaded image exceeds the size limit.	The module returned `"FILE_SIZE_INVALI`	Pass

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ID	Action	Expected Result	Actual Result	Result
IMGVAL-UT4	Validate request with a corrupted image file.	The module returns `"UNAUTHORIZED_ACCESS_OR_CORRUPTED"` to indicate that the uploaded image cannot be accessed.	The module returned `"UNAUTHORIZED_ACCESS_OR_CORRUPTED"`.	Pass
IMGVAL-UT5	Validate request with a valid image file.	The module returns `"Success"` to indicate that the uploaded image meets system requirements and has been successfully uploaded.	The module returned `"Success"`.	Pass

7.2 BackendController Module

Table 9 summarizes the unit testing results for the BackendController module.

Table 9: BackendController Module Unit Testing Evaluation

ID	Action	Expected Result	Actual Result	Result
BCONT-UT1	Send image validation request with valid image data through the backend controller.	The backend controller forwards the image to the Image Validation module and returns the validation results.	The backend controller correctly returns `Success` after calling the image validation module.	Pass
BCONT-UT2	Send alt text generation request with image data and a valid session ID through the backend controller.	The backend controller forwards the request to the Alt Text Generation module and returns the generated alt text.	The backend controller correctly returns the generated alt text of `"Generated alt text"`.	Pass
BCONT-UT3	Send alt text history retrieval request with a valid session ID through the backend controller.	The backend controller forwards the session ID to the Alt Text History module and returns the retrieved history.	The backend controller correctly returns the retrieved history of `["Alt text 1", "Alt text 2"]`.	Pass
BCONT-UT4	Send alt text editing request through the backend controller.	The backend controller forwards the request to the Edit Alt Text module and returns the result.	The backend controller correctly returns the editing result of `"Success"`.	Pass

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ID	Action	Expected Result	Actual Result	Result
EDIT-UT1	Submit a valid edit request with valid entry ID, edited alt text and session ID.	The alt text is updated with the edited text and returns a success message.	The service correctly returns `"Success"` after calling the service to update the history data.	Pass
EDIT-UT2	Submit a valid edit request but simulate a failed save to the database.	The service returns and error indicating the change cannot be saved.	The service correctly returns an error message of `"UNABLE_TO_SAVE"` and does not call the service to update the history data.	Pass
EDIT-UT3	Submit an edit request with the `entry_id` field missing.	The service detects the missing field and returns an error indicating the missing field and does not attempt to update the alt text.	The service correctly returns an error message of `"INVALID_REQUEST"` and does not call the service to update the history data.	Pass
EDIT-UT4	Submit an edit request with the `edited_alt_text` field missing.	The service detects the missing field and returns an error indicating the missing field and does not attempt to update the alt text.	The service correctly returns an error message of `"INVALID_REQUEST"` and does not call the service to update the history data.	Pass

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ID	Action	Expected Result	Actual Result	Result
EDIT-UT5	Submit an edit request but simulate an exception being thrown by the Supabase client.	The service catches the exception and returns an error indicating the failure and does not attempt to update the alt text.	The service correctly returns an error message of `"UNABLE_TO_SAVE"` after the exception is caught	Pass
GEN-UT1	Submit alt text with extra spaces for post-processing.	The extra spaces are removed from the alt text and its returned.	The function correctly removes extra spaces from the alt text.	Pass
GEN-UT2	Submit alt text with repeated words for post-processing.	The repeated words are removed from the alt text and it is returned.	The function correctly removes repeated words from the alt text.	Pass
GEN-UT3	Submit alt text starting with a lowercase letter for post-processing.	The first letter is capitalized and it is returned.	The function correctly capitalizes the first letter of the alt text.	Pass
GEN-UT4	Submit alt text with a missing period for post-processing.	A period is added to the end of the alt text and it is returned.	The function correctly adds a period to the end of the alt text.	Pass

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ID	Action	Expected Result	Actual Result	Result
GEN-UT5	Submit alt text with multiple sentences beginning with lowercase letters for post-processing.	Each sentence is capitalized and the text is returned.	The function correctly capitalizes each sentence in the alt text.	Pass
GEN-UT6	Trigger the AI model for AI alt text generation and simulate no text is returned	The function returns no alt text and no update to history is completed.	The function correctly returns `(None, None)`	Pass
GEN-UT7	Trigger the AI model for AI alt text generation and simulate alt text is returned	The function returns the generated alt text and updates the history.	The function correctly returns the generated alt text and updates the history.	Pass

7.3 AuthenticationService Module

Table 10 summarizes the unit testing results for the AuthenticationService module.

Table 10: AuthenticationService Module Unit Testing Evaluation

ID	Action	Expected Result	Actual Result	Result
AUTH-UT1	Send a request with a valid email and password to the AuthenticationService.	The authentication service successfully creates a new user account and returns the user ID and access token.	The authentication service correctly returns the user ID and access token, i.e., {`"user_id": "test_user_id", "access_token": "test_access_toke"`}	Pass
AUTH-UT2	Send request with invalid email format to the AuthenticationService	The authentication service raises a `ValueError` with the message "Email must be a @mcmaster.ca address". The supabase client is not called.	The authentication service correctly raises a `ValueError` with the message "Email must be a @mcmaster.ca address" and the supabase client is not called.	Pass
AUTH-UT3	Send a request with a password shorter than 8 characters to the AuthenticationService.	The authentication service raises a `ValueError` with the message "Password must be at least 8 characters". The Supabase client is not called.	The authentication service correctly raises a `ValueError` with the message "Password must be at least 8 characters" and the Supabase client is not called.	Pass

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ID	Action	Expected Result	Actual Result	Result
AUTH-UT4	Send a request with empty email and password to the AuthenticationService.	The authentication service raises a `ValueError` with the message "Email must be a @mcmaster.ca address". The Supabase client is not called.	The authentication service correctly raises a `ValueError` with the message "Email must be a @mcmaster.ca address" and the Supabase client is not called.	Pass
AUTH-UT5	Send a request with `STUDENT@mcmaster.ca` as the email to the AuthenticationService.	The authentication service correctly converts the email to lowercase before processing.	The authentication service correctly converts the email to lowercase before processing. The Supabase client is called with `student@mcmaster.ca`	Pass
AUTH-UT6	Send a request with an existing email to the AuthenticationService.	The authentication service raises a `ValueError` with the message "Email is already in use". A verification email is not sent to the user	No error was raised and the user received a Verification email.	Fail

Analysis of Failed Tests:

- **AUTH-UT6:** The failure in AUTH-UT6 occurred because the system al-

lowed account creation using an email that already exists, instead of raising an error as expected. This indicates that duplicate account prevention was not properly enforced at the application level. The issue is likely due to reliance on the external authentication service (Supabase), which may not restrict duplicate sign-up attempts by default, or the absence of an explicit validation check before account creation. This reflects an implementation gap rather than a requirement ambiguity or code quality issue. Future improvements should include enforcing unique user constraints and validating existing accounts prior to registration.

7.4 MainScreen Module

Table 11 summarizes the unit testing results for the MainScreen module.

Table 11: MainScreen Unit Testing Evaluation

ID	Action	Expected Result	Actual Result	Result
MAIN-UT1	Render HomeScreen page	Homepage title is displayed	Homepage title displayed successfully	Pass
MAIN-UT2	Render HomeScreen subtitle	Homepage subtitle describing system purpose is visible	Subtitle displayed correctly	Pass
MAIN-UT3	Display signup button on homepage	Signup button is visible for new users	Signup button rendered successfully	Pass
MAIN-UT4	Display login button on homepage	Login button is visible for returning users	Login button rendered successfully	Pass
MAIN-UT5	Render UploadScreen title	Upload page title is visible	Title displayed correctly	Pass

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ID	Action	Expected Result	Actual Result	Result
MAIN-UT6	Render UploadScreen subtitle	Upload page subtitle guiding users is visible	Subtitle displayed correctly	Pass
MAIN-UT7	Verify upload instructions accessibility	Upload instructions accessible through ARIA label	ARIA label detected successfully	Pass
MAIN-UT8	Verify upload input accessibility label	Upload input field contains accessible label	Label detected successfully	Pass
MAIN-UT9	Display Choose File button	Choose File button is visible for uploading images	Button displayed successfully	Pass
MAIN-UT10	Upload valid image file	Image successfully uploaded and backend request triggered	Image uploaded and confirmation displayed	Pass
MAIN-UT11	Upload image using drag-and-drop	Image uploaded using drag-and-drop functionality	Drag-and-drop upload completed successfully	Pass
MAIN-UT12	Remove uploaded image	Uploaded image removed from interface	Image removed successfully	Pass
MAIN-UT13	Generate alt text after image upload	Generated alt text displayed in output field	Alt text generated and displayed successfully	Pass

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ID	Action	Expected Result	Actual Result	Result
MAIN-UT14	Upload new image after generation	Previous alt text resets when new image uploaded	Alt text reset confirmed	Pass
MAIN-UT15	Copy generated alt text	Alt text copied to clipboard successfully	Clipboard copy function executed successfully	Pass
MAIN-UT16	Keyboard accessibility for file upload	Pressing Enter opens file selection dialog	File input triggered successfully	Pass
MAIN-UT17	Handle image upload error	Error message displayed when upload fails	Error message displayed correctly	Pass
MAIN-UT18	Handle alt text generation failure	Error message displayed when alt text generation fails	Error message displayed correctly	Pass

7.5 ShowHistory Module

Table 12 summarizes the unit testing results for the ShowHistory module.

Table 12: ShowHistory Unit Testing Evaluation

ID	Action	Expected Result	Actual Result	Result
HIST-UT1	Render ShowHistory page	History page title and navigation button are displayed	Title and back button displayed successfully	Pass
HIST-UT2	Display history entries	Uploaded image history entries are listed on the page	History entries rendered successfully	Pass

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ID	Action	Expected Result	Actual Result	Result
HIST-UT3	View history entry preview	Clicking a history entry displays the image preview and alt text	Preview image and alt text displayed correctly	Pass
HIST-UT4	Navigate back to upload screen	Clicking the back button navigates to the upload page	Navigation executed successfully	Pass

7.6 LoginScreen Module

Table 13 summarizes the unit testing results for the LoginScreen module.

Table 13: LoginScreen Unit Testing Evaluation

ID	Action	Expected Result	Actual Result	Result
LOGIN-UT1	Render login interface	Login page title and input fields are visible	Login interface rendered successfully	Pass
LOGIN-UT2	Submit login with empty fields	System rejects login attempt and displays validation error	Validation error displayed successfully	Pass
LOGIN-UT3	Enter email and password credentials	User can type and update values in email and password fields	Input fields accept user input correctly	Pass

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ID	Action	Expected Result	Actual Result	Result
LOGIN-UT4	Login with valid credentials	System sends authentication request and redirects user to upload page	Authentication request successful and navigation executed	Pass
LOGIN-UT5	Login with invalid credentials	System displays authentication error message	Error message displayed correctly	Pass
LOGIN-UT6	Network failure during login	System displays network error message	Network error message displayed successfully	Pass
LOGIN-UT7	Render signup form interface	Signup form fields and instructions are displayed	Signup interface rendered successfully	Pass
LOGIN-UT8	Signup validation with missing email	System displays validation error when email is not provided	Validation error displayed successfully	Pass
LOGIN-UT9	Signup validation for short password	System rejects passwords shorter than minimum length	Password validation message displayed	Pass
LOGIN-UT10	Signup validation for mismatched passwords	System rejects mismatched password entries	Mismatch validation error displayed	Pass
LOGIN-UT11	Send verification code	Verification code request sent to backend and OTP field enabled	Verification request executed successfully	Pass

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ID	Action	Expected Result	Actual Result	Result
LOGIN-UT12	Signup attempt without OTP	System prevents signup and displays OTP required error	OTP validation error displayed	Pass
LOGIN-UT13	Complete successful signup	System verifies OTP and completes account registration	Signup process completed successfully	Pass
LOGIN-UT14	RequireAuth loading state	Protected content not rendered until authentication check completes	Content hidden during authentication check	Pass
LOGIN-UT15	RequireAuth authenticated access	Authenticated users can access protected content	Protected content rendered successfully	Pass
LOGIN-UT16	RequireAuth unauthorized access	Unauthenticated users are redirected to login page	Unauthorized access prevented and redirect executed	Pass

8 Manual Testing

This section summarizes the manual tests carried out for the major system modules and presents the observed results.

8.1 UserInterfaceInteractions Module

The following are the manual tests for this module.

UII-MT 1. **Initial State:** User is logged in and the main screen is displayed.
Input: User selects and uploads an image file using the upload button.
Test Steps:

(a) Click the Choose File button.

(b) Select a valid image file (.png or .jpeg).

(c) Submit the upload.

Expected Results: The system accepts the image and sends it to the backend for processing. The uploaded image appears in the interface and alt-text generation begins.
Failure Condition: The image fails to upload, the UI freezes, or the system does not start alt-text generation.

UII-MT 2. **Initial State:** User is logged in and the main screen is displayed.
Input: User attempts to upload an unsupported file type or file exceeding the maximum size.
Test Steps:

(a) Click the Choose File button.

(b) Select an unsupported file (e.g., .gif, .pdf) or a file larger than the allowed size.

(c) Submit the upload.

Expected Results: The system rejects the file and displays an error message such as "File type not supported" or "File too large."
Failure Condition: The system accepts unsupported files or fails to display an error message.

UII-MT 3. **Initial State:** Generated alt text is displayed on the screen.
Input: User presses the Copy button to copy the generated alt text.
Test Steps:

(a) Generate alt text for an uploaded image.

(b) Click the Copy Alt Text button.

(c) Paste the clipboard contents into another application (e.g., a text editor).

Expected Results: The copied text matches the generated alt text exactly.
Failure Condition: The clipboard is empty or the copied text is incomplete or incorrect.

UII-MT 4. **Initial State:** Generated alt text is visible and editable on the screen.
Input: User edits the alt text and saves the changes.
Test Steps:

(a) Modify the alt text by adding, deleting, or changing words.

(b) Click the Save Edits button.

Expected Results: The updated alt text is displayed and stored correctly without formatting issues.
Failure Condition: The edits are not saved, the text reverts to its original version, or formatting errors occur.

UII-MT 5. **Initial State:** Login screen is displayed.
Input: User enters valid credentials and attempts to sign in.
Test Steps:

(a) Enter valid login credentials.

(b) Click the Sign In button.

Expected Results: The user is authenticated and redirected to the main system interface.
Failure Condition: Valid credentials are rejected or no response occurs after clicking Sign In.

UII-MT 6. **Initial State:** User is logged into the system.
Input: User presses the Sign Out button.
Test Steps:

(a) Click the Sign Out button.

Expected Results: The user session ends and the system redirects the user to the login screen.
Failure Condition: The user remains logged in or can continue accessing the system without logging in again.

The following table shows the evaluation of the userinterfaceinteractions module's manual tests. Table 14 summarizes the results.

Table 14: UserInterfaceInterfactions Manual Testing Evaluation

ID	Action	Expected Result	Actual Result	Result
UII-MT1	Upload a valid image file (.png or .jpeg) using the Choose File button	Image is accepted and alt-text generation begins	Image uploaded successfully and alt-text generation initiated	Pass
UII-MT2	Upload an unsupported file type or file exceeding the allowed size	System rejects file and displays an appropriate error message	Unsupported file was rejected and an error message was displayed	Pass
UII-MT3	Copy generated alt text using the Copy Alt Text button	Alt text is copied to clipboard accurately	Alt text copied successfully and matched the generated output when pasted	Pass
UII-MT4	Edit generated alt text and save changes	System saves and displays updated alt text without formatting issues	Edits were saved correctly and updated text was displayed	Pass
UII-MT5	Enter valid credentials and click Sign In	User is authenticated and redirected to the main interface	Login successful and user was redirected to the main screen	Pass
UII-MT6	Click the Sign Out button	User session ends and login screen is displayed	User session ended and system returned to the login screen	Pass

8.2 UserInterfaceAccessibility Module

The following are the manual tests for this module.

UIA-MT 1. **Initial State:** A system screen such as the login screen, main screen, or history screen is loaded.
Input: The tester inspects the rendered page structure using browser accessibility tools or screen reader landmarks.
Test Steps:

(a) Open a system screen.

(b) Inspect the page structure using accessibility inspector or a screen reader landmark list.

(c) Verify that the main content area is identified as the page's main landmark.

Expected Results: The screen contains a valid main landmark (`<main>` or equivalent) for the primary content area.
Failure Condition: No main landmark is present, or the primary content is not identified correctly.

UIA-MT 2. **Initial State:** The login screen or another screen (main or history screen)has just loaded.
Input: The tester loads the screen and checks where keyboard focus is placed.
Test Steps:

(a) Open the login screen.

(b) Do not click the mouse.

(c) Observe which element receives focus automatically.

(d) Press `Tab` to continue keyboard navigation if needed.

Expected Results: Keyboard focus is moved to the intended first interactive element, such as the username field on the login screen.
Failure Condition: Focus is missing, placed on a non-interactive element, or placed on the wrong control.

UIA-MT 3. **Initial State:** A screen reader is enabled and the user performs an action that changes the UI.
Input: Trigger a UI change such as loading the login screen, generating alt text, or navigating to history.
Test Steps:

(a) Enable a screen reader such as NVDA, JAWS, or VoiceOver.

(b) Perform an action that changes the interface.

(c) Listen for the announcement made by the system.

Expected Results: A clear announcement describing the UI change is read by the screen reader through the live region.
Failure Condition: No announcement is made, the message is incomplete, or the announcement is unclear.

UIA-MT 4. **Initial State:** A screen containing multiple interactive elements is open.
Input: The tester uses only the keyboard to navigate within the interface.
Test Steps:

(a) Open a screen with interactive controls.

(b) Use `tab` to move between elements where applicable.

(c) Use `Enter` or `Space` to activate the selected control.

Expected Results: Keyboard navigation works correctly within the defined scope. Focus moves predictably and activation occurs without changing application state unexpectedly.
Failure Condition: Tab key do not work, focus movement is inconsistent, or activation fails.

The following table shows the evaluation of the userinterfaceaccessibility module's manual tests. Table 15 summarizes the results.

Table 15: UserInterfaceAccessibility Manual Testing Evaluation

ID	Action	Expected Result	Actual Result	Result
UIA-MT1	Verify that the loaded screen contains a valid main landmark for primary content	Main content area is identified by a valid main landmark	Main landmark was not present and did not identify the primary content area	Fail
UIA-MT2	Load the screen and check automatic keyboard focus placement	Focus is placed on the intended first interactive element	Initial focus was correctly placed on the expected interactive control	Pass
UIA-MT3	Trigger a UI change while a screen reader is enabled	Screen reader announces the interface change clearly	Announcement was read clearly by the screen reader after any UI update	Pass
UIA-MT4	Use arrow keys, Home/End, and Enter/Space to navigate and activate controls	Keyboard navigation works correctly within the interface scope	Keyboard navigation and activation worked as expected without unexpected state changes	Pass

Analysis of Failed Tests:

- **UIA-MT1:** These failures were due to missing UI elements (e.g., main landmark) and incomplete implementation of accessibility documentation and best practices. The issues stem from implementation gaps and evolving requirements rather than fundamental design flaws. Automated tools helped identify these gaps for future correction.

8.3 DataPreProcess Module

The following are the manual tests for this module.

DP-MT 1. **Initial State:** The preprocessing function is available and ready to process dataset examples.
Input: A dataset example containing valid image paths and modified alt text.
Test Steps:

(a) Provide a dataset example containing valid image paths and alt text.

(b) Run the preprocess function.

(c) Inspect the returned encoded output.

Expected Results: The function successfully loads images and text and returns encoded tensors containing image and text representations.
Failure Condition: The preprocessing function fails to return encoded outputs or raises an unexpected error.

DP-MT 2. **Initial State:** The preprocessing function is ready to process input images.
Input: Image files that may not originally be in RGB format.
Test Steps:

(a) Provide image paths to the preprocess function.

(b) Run preprocessing.

(c) Inspect the processed images.

Expected Results: All images are converted to RGB format before being passed to the processor.
Failure Condition: Images remain in their original format and are not converted to RGB.

DP-MT 3. **Initial State:** The preprocessing function receives alt text label with leading or trailing whitespace.
Input: Alt text label containing extra whitespace characters.
Test Steps:

(a) Provide alt text label with leading or trailing spaces.

(b) Run preprocessing.

(c) Inspect the processed text values.

Expected Results: Leading and trailing whitespace is removed from the alt text.
Failure Condition: Whitespace remains in the processed alt text.

DP-MT 4. **Initial State:** The preprocessing function is ready to process dataset examples.
Input: A dataset example containing an invalid or unreadable image path.
Test Steps:

(a) Provide a dataset example containing an invalid image path.

(b) Run the preprocess function.

(c) Observe the system response.

Expected Results: The function raises an error indicating that the image cannot be opened or read.
Failure Condition: The function silently proceeds or produces invalid encoded output.
eeds the maximum length.

The following table shows the evaluation of the datapreprocess module's manual tests. Table 16 summarizes the results.

Table 16: DataPreprocess Manual Testing Evaluation

ID	Action	Expected Result	Actual Result	Result
DPP-MT1	Process valid image paths and alt text labels through the preprocess function	Valid images and text are encoded successfully	Images and alt text were processed successfully and encoded tensors were returned	Pass

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ID	Action	Expected Result	Actual Result	Result
DPP-MT2	Provide image paths and run preprocessing to verify RGB conversion	Images are converted to RGB format before encoding	Images were successfully converted to RGB format before processing	Pass
DPP-MT3	Provide alt text containing leading or trailing whitespace	Whitespace is removed from alt text	Alt text whitespace was successfully stripped before encoding	Pass
DPP-MT4	Provide an invalid or unreadable image path to preprocessing	Function raises an error indicating the image cannot be opened	Invalid image path correctly triggered an error during preprocessing	Pass

8.4 ModelOutput Module

Table 17 summarizes the manual testing result for the ModelOutput module.

The following are the manual tests for this module.

MO-MT 1. **Initial State:** The system has successfully generated alt-text for an uploaded image.

Input: The backend returns the generated caption to be displayed to the user.

Test Steps:

(a) Upload a valid diagram image.

(b) Wait for the alt-text generation process to complete.

(c) Observe the output displayed on the user interface.

Expected Results: The generated alt-text is correctly formatted and displayed to the user through the interface.

Failure Condition: The generated alt-text is missing, incorrectly formatted, or not displayed to the user.

Table 17: Manual Testing Evaluation for ModelOutput Module

Test ID	Module	Description	Actual Result	Result
MO-MT1	ModelOutput Module	Verify generated alt-text is correctly formatted and displayed to the user.	All assertions pass	Pass

8.5 AIModelTraining Module

Table 18 summarizes the manual testing result for the AIModelTraining module.

The following are the manual tests for this module.

AI-MT 1. **Initial State:** The training dataset and configuration parameters are available.

Input: The model training process is initiated using the dataset.

Test Steps:

(a) Start the training pipeline.

(b) Load the dataset used for training the model.

(c) Monitor the training process for completion.

Expected Results: The model successfully completes training and produces a trained model artifact used for caption generation.

Failure Condition: Training fails to complete, produces errors, or does not generate a usable trained model.

Table 18: Manual Testing Evaluation for AIModelTraining Module

Test ID	Module	Description	Actual Result	Result
AI-MT1	AIModelTraining Module	Verify that the model training process successfully produces a trained model.	All assertions pass	Pass

8.6 CaptionGeneration Module

Table 19 summarizes the manual testing result for the CaptionGeneration module.

The following are the manual tests for this module.

CG-MT 1. **Initial State:** A trained model exists and a valid image has been uploaded.

Input: The system processes the uploaded image using the trained model.

Test Steps:

(a) Upload a valid diagram image.
(b) Submit the image for caption generation.
(c) Wait for the caption generation process.

Expected Results: The system generates descriptive alt-text that reflects the content of the image.

Failure Condition: The system fails to generate alt-text, produces irrelevant text, or returns an error.

Table 19: Manual Testing Evaluation for CaptionGeneration Module

Test ID	Module	Description	Actual Result	Result
CG-MT1	CaptionGeneration Module	Verify that the system generates descriptive alt-text for uploaded images.	All assertions pass	Pass

8.7 InternalMetricCompliance Module

Table 20 summarizes the manual testing result for the InternalMetricCompliance module.

The following are the manual tests for this module.

IMC-MT 1. **Initial State:** A caption has been generated by the Caption-Generation module.

Input: The generated caption is evaluated using internal quality metrics.

Test Steps:
(a) Generate alt-text from an uploaded image.
(b) Allow the system to run internal metric validation checks.
(c) Observe whether the caption passes the internal quality checks.

Expected Results: The caption is evaluated against internal metrics and confirmed to meet clarity and quality standards.

Failure Condition: The caption fails validation checks or the system does not evaluate the generated caption.

Table 20: Manual Testing Evaluation for InternalMetricCompliance Module

Test ID	Module	Description	Actual Result	Result
IMC-MT1	InternalMetricCompliance Module	Verify that generated captions satisfy internal quality evaluation metrics.	All assertions pass	Pass

9 Changes Due to Testing

During the internal testing phase, several changes were made to the tool based on feedback from the TA, professor, supervisor and from Team 22. A brief description of each change and its implementation is listed below:

9.1 Fixing Session Timeout Handling

During testing, it was observed that user sessions did not expire after the defined `SESSION_TIMEOUT` period of inactivity. This posed a potential security risk, as inactive users could remain logged into the system indefinitely. To address this issue, the session management logic was updated to properly track user inactivity and automatically terminate sessions once the timeout threshold was reached. The system now redirects users to the login page and requires re-authentication after inactivity exceeds `SESSION_TIMEOUT`. This change improves system security and ensures compliance with authentication and session management requirements.

9.2 Narrowing the Scope

While the team had the ambition to make a STEM-alt text generator, this proved to be very expensive in terms of time needed for training and constraints on available computers for AI model tuning. To make up for this, Team 22 narrowed the scope to only "Physics" based diagrams, with a plan to further narrow the image types to a select list. This change would help our AI model learn better, contextual text and in turn produce higher quality, specific alt text that is usable for several stakeholders like students and teachers.

9.3 Shortening Alt-Text Generation Time

The AI model initially took five minutes or more to generate text for one image. After talking to our supervisor, Team 22 was able to decide on a faster model that provided mediocre alt-text generation while adding in indicators for user navigations and processes. We were able to cut down from approximately five minutes per image to 1-2 minutes per image, which significantly reduced user frustration.

9.4 Clearer User Navigation

During testing, Team 22 had found that some user navigation processes were not as clear to a screen-reader user or someone with a visual impairment. Examples

include no descriptive 'aria' labels or a misplaced page navigation. To fix this, we tried testing both automatically and manually in several ways including testing the solution with just the keyboard and voice assist. This significantly improved our solution, reduced tester frustration and gave users a clear indication of what was going on behind the scenes.

9.5 Future Work

While our testing showed us many faults in our solution, not all of them could be fixed. For future revisions, Team 22 plans to further increase alt-text quality by adding more specificity and context to generation (using AI hypertuning methods). Conducting user testing has given us sufficient data to know what features and system modules to improve for better usability.

10 Automated Testing

This section describes the automated testing approaches used to evaluate the system and support overall verification.

10.1 Accessibility Checker - Axe

The accessibility checks were done using a third-party website, Axe, which helped Team 22 understand gaps in user navigation for people with disabilities. Figure 1 shows an example of a generated accessibility report.

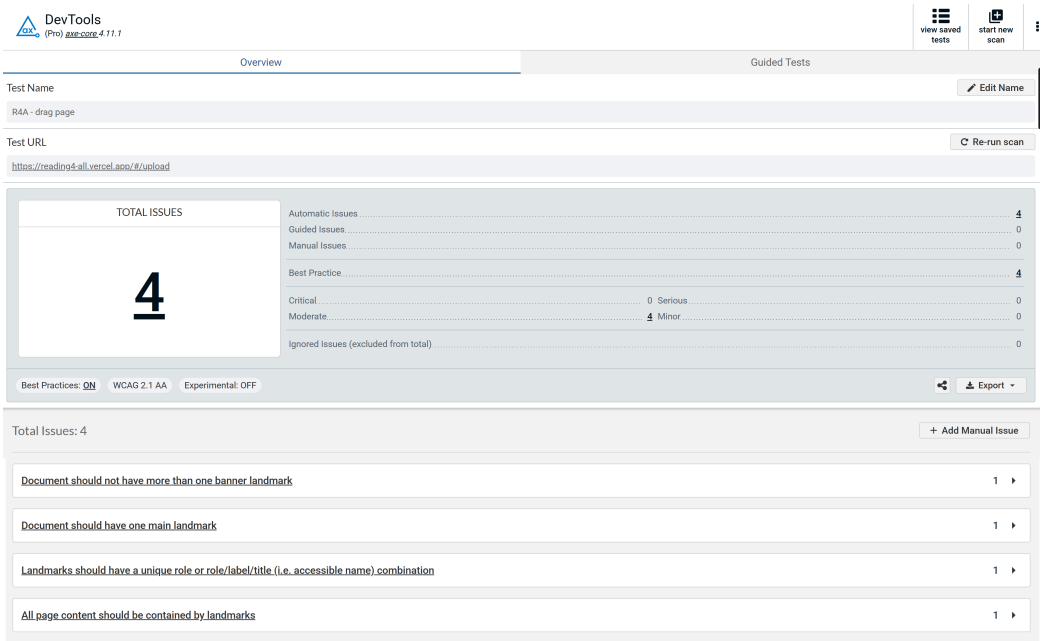

Figure 1: Automated Report for "Upload Page" in the Reading4All System

Reports like these made it easier for the team to identify accessibility oversights and best practices we might have missed and fix them. Table 21 summarizes the Axe audit findings across the three main application pages.

Table 21: Summary of Axe Accessibility Audit Results by Page

Page	**Critical**	**Serious**	**Moderate**	**Notes**
Login / Signup	0	0	1	Missing `autocomplete` attribute on password field
Upload (Main)	0	1	2	Missing `<main>` landmark (serious); duplicate `id` on rendered list items (moderate)
History	0	0	1	Image previews lack `alt` attribute in some dynamic renders
Total	**0**	**1**	**4**	No critical violations detected

The missing `<main>` landmark (Upload page, serious) has since been corrected as described in the analysis of NFR-ST22. The remaining moderate issues are tracked as known defects for the next release. No critical or serious violations remain in the most recent build.

10.2 CI/CD

Continuous Integration and Continuous Deployment (CI/CD) practices were used to ensure that changes to the Reading4All system could be integrated and tested efficiently. Whenever updates were pushed to the repository, automated workflows were triggered to verify that the system built successfully and that core components continued to function as expected.

These automated checks helped the team detect issues early in the development process and prevented unstable code from being merged into the main branch. By incorporating CI/CD into the development workflow, Team 22 was able to maintain code quality, streamline collaboration, and ensure that new features or fixes did not unintentionally break existing functionality.

Additionally, the CI/CD pipeline supported reproducibility by ensuring that the same build and testing procedures were executed consistently for each update. This contributed to a more stable and reliable development process for the Reading4All system.

10.3 Mutation Testing

Mutation testing was originally proposed in the VnV Plan as a method to evaluate the effectiveness of unit tests by introducing small faults (mutations) into the code and verifying whether the existing tests could detect them.

However, mutation testing was not fully implemented in this project due to time constraints and the additional setup complexity required for integrating mutation testing tools with both the React frontend and Python backend environments. Additionally, the team prioritized functional correctness, accessibility testing, and integration validation, which were more critical to the system's goals.

Despite not conducting formal mutation testing, test effectiveness was indirectly assessed through code coverage analysis and the ability of unit tests to detect real defects during development. Future work could include incorporating mutation testing tools to further evaluate and strengthen the robustness of the test suite.

11 Trace to Requirements

The following table (Table 22) shows the relationship between the tests performed and the requirements defined in the SRS (2025).
Note: Each test ID range maps to the SRS requirements it validates. For a matrix-style view showing individual test-to-requirement coverage, see Appendix 13.

Table 22: Traceability Between Tests and Requirements

Test ID	**SRS Requirement(s) Satisfied**
IMGVAL-UT1 – IMGVAL-UT5	FR 1, MD-SL 3, SR-IM 1
BCONT-UT1 – BCONT-UT4	FR 1, FR 2, FR 4, FR 5
AUTH-UT1 – AUTH-UT6	FR 6, SR-AR 1, SR-IM 2
MAIN-UT1 – MAIN-UT18	FR 1, FR 2, FR 4, UHR-EUR 1, UHR-EUR 2
HIST-UT1 – HIST-UT4	FR 5, UHR-EUR 2
LOGIN-UT1 – LOGIN-UT16	FR 6, SR-AR 1, SR-PR 1
UII-MT1 – UII-MT6	FR 1, FR 2, FR 4, FR 6, UHR-PIR 1
UIA-MT1 – UIA-MT4	LFR-AR 1, LFR-AR 3, LFR-AR 4, UHR-AR 2
DPP-MT1 – DPP-MT4	PR-SL 1, PR-SL 2, PR-PAR 1
MO-MT1	FR 2, PR-PAR 2
AI-MT1	PR-PAR 3
CG-MT1	FR 2, PR-PAR 1
IMC-MT1	PR-PAR 1, PR-PAR 2, PR-PAR 3
FR-ST1 – FR-ST7	FR 1 – FR 6
NFR-ST1 – NFR-ST26	LFR-AR 1 – LFR-AR 4, UHR-EUR 1 – UHR-EUR 4, PR-SL 1 – PR-SL 2, SR-AR 1, SR-PR 1, OER 1

12 Trace to Modules

The following table (Table 23) shows the relationship between the tests performed and the modules defined in the Module Guide. Each test verifies the correct

functionality of the corresponding module in the Reading4All system. Note: Each test ID range maps to the SRS requirements it validates. For a matrix-style view showing individual test-to-requirement coverage, see Appendix 13.

Table 23: Traceability Between Tests and Modules

Test ID	**Module**
IMGVAL-UT1 – IMGVAL-UT5	ImageValidation Module
BCONT-UT1 – BCONT-UT4	BackendController Module
AUTH-UT1 – AUTH-UT6	AuthenticationService Module
MAIN-UT1 – MAIN-UT18	MainScreen Module
HIST-UT1 – HIST-UT4	ShowHistory Module
LOGIN-UT1 – LOGIN-UT16	LoginScreen Module
UII-MT1 – UII-MT6	UserInterfaceInteractions Module
UIA-MT1 – UIA-MT4	UserInterfaceAccessibility Module
DPP-MT1 – DPP-MT4	DataPreProcess Module
MO-MT1	ModelOutput Module
AI-MT1	AIModelTraining Module
CG-MT1	CaptionGeneration Module
IMC-MT1	InternalMetricCompliance Module
FR-ST1 – FR-ST7	System Functional Tests (cross-module verification)
NFR-ST1 – NFR-ST26	System Nonfunctional Tests (cross-module verification)

13 Code Coverage Metrics

The following analyzes the code coverage metrics for the React frontend of the system. The analysis is based on the coverage data generated using Jest and React Testing Library as shown in Figure 2. Code coverage is a measure of how well the codebase is tested, and it helps identify areas that may require additional testing.

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	**79.27**	**70.52**	**72.05**	**79.93**	
src	0	100	0	0	
App.js	0	100	0	0	12
index.js	0	100	100	0	6-7
src/components	0	0	0	0	
Header.jsx	0	0	0	0	6-42
src/pages	100	83.33	100	100	
RequireAuth.jsx	100	83.33	100	100	22
src/pages/HistoryScreen	87.09	80	83.33	87.09	
HistoryPage.jsx	87.09	80	83.33	87.09	22-26,35,55
src/pages/HomeScreen	50	100	33.33	50	
homeScreen.jsx	50	100	33.33	50	17-24
src/pages/LogInScreen	100	92.85	100	100	
login.jsx	100	92.85	100	100	33
src/pages/ShowHistoryScreen	88.88	50	80	88.88	
showHistory.jsx	88.88	50	80	88.88	38
src/pages/SignUpScreen	83.33	74.28	90	83.33	
SignUpScreen.jsx	83.33	74.28	90	83.33	42,71,111-112,118,130-131,148-150,156,242
src/pages/UploadScreen	77.03	72.28	65.21	78.78	
UploadScreen.jsx	77.03	72.28	65.21	78.78	31-32,36,61,65,71-102,117,128-129,134,158,175,192,222-226,321-419

```
Test Suites: 7 passed, 7 total
Tests:       45 passed, 45 total
Snapshots:   0 total
Time:        1.543 s, estimated 2 s
Ran all test suites.
```

Figure 2: Coverage Report of the React Frontend

The following analyzes the code coverage for the backend of the system. This analysis was performed using `pytest-cov` and shows how key features such as authentication, image validation, session history of the system are unit tested. Furthermore, files such as `alt_text_history.py` and `gen_alt_text.py` have lower unit test coverage due to their reliance on Supabase database services. As a result, their functionality was additionally tested manually to ensure correct integration with Supabase and to ensure the expected behavior. Please refer to the Figure 3 for more detail and insight.

```
================================ tests coverage ================================
_______________ coverage: platform darwin, python 3.12.6-final-0 _______________

Name                                          Stmts   Miss  Cover
---------------------------------------------------------------
api/__init__.py                                   0      0   100%
api/admin.py                                      0      0   100%
api/apps.py                                       3      3     0%
api/management/__init__.py                        0      0   100%
api/management/commands/__init__.py               0      0   100%
api/management/commands/test_supabase.py          9      4    56%
api/middleware.py                                22     22     0%
api/migrations/0001_initial.py                    6      6     0%
api/migrations/__init__.py                        0      0   100%
api/models.py                                    14     14     0%
api/tests.py                                      0      0   100%
backend/__init__.py                               0      0   100%
backend/asgi.py                                   4      4     0%
backend/settings.py                              29     29     0%
backend/urls.py                                   3      3     0%
backend/wsgi.py                                   5      5     0%
controller/backend_controller.py                 23      1    96%
databases/connect_supabase.py                    18     10    44%
manage.py                                        11     11     0%
services/alt_text_history.py                     26     10    62%
services/auth_service.py                         16      0   100%
services/edit_alt_text.py                        18      0   100%
services/gen_alt_text.py                         67     19    72%
services/image_validation.py                     18      0   100%
tests/test_alt_text_history.py                   62      0   100%
tests/test_auth_service.py                       71      5    93%
tests/test_backend_controller.py                 31      0   100%
tests/test_edit_alt_text.py                      51      0   100%
tests/test_gen_alt_text.py                       64      0   100%
tests/test_image_validation.py                   43      0   100%
---------------------------------------------------------------
TOTAL                                           614    146    76%
```

Figure 3: Coverage Report of the Backend

References

Software requirements specification for bridging gaps: Ai for diagram accessibility, 2025. URL https://github.com/4G06-CAPSTONE-2025/Reading4All/blob/main/docs/SRS-Volere/SRS.pdf.

Appendix — Traceability Matrix

Table 24 provides a condensed matrix mapping selected test IDs to key functional and non-functional requirements.

Table 24: Test-to-Requirement Traceability Matrix (Selected Tests)

Test ID	**FR1**	**FR2**	**FR4**	**FR5**	**FR6**	**LFR-AR1**	**SR-AR1**
FR-ST1	✓						
FR-ST2		✓					
FR-ST3		✓				✓	
FR-ST4			✓				
FR-ST5				✓			
FR-ST6					✓		✓
FR-ST7					✓		✓
NFR-ST1						✓	
NFR-ST5		✓				✓	
NFR-ST12					✓		✓

Appendix — Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Reflection.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response: None.

Team Reflection

1. Which parts of this document stemmed from speaking to your client(s) or a proxy (e.g. your peers)? Which ones were not, and why?

   One aspect that stemmed from speaking with our supervisor is the importance of manual testing over automated testing, specifically the user interface. Our supervisor emphasized the importance of ensuring our interface is accessible and this is highlighted through our various nun-functional tests. We tried our best to put ourselves in the users shoes and understand any difficulties or issues they might face while using the Reading4All application. Furthermore, determining which aspects of the system to unit test verses manually test was determined through discussions with our TA supervisor. This was an aspect we were initialling unsure about; however, after discussions we determined that certain aspects such as the system as a whole, the accessibility, and alt text generation might be better manually tested. The Reading4All system implementation was mainly derived from our previous learnings through courses and industry standards, rather than from our clients and peers.

2. In what ways was the Verification and Validation (VnV) Plan different from the activities that were actually conducted for VnV? If there were differences, what changes required the modification in the plan? Why did these changes occur? Would you be able to anticipate these changes in future projects? If there weren't any differences, how was your team able to clearly predict a feasible amount of effort and the right tasks needed to build the evidence

that demonstrates the required quality? (It is expected that most teams will have had to deviate from their original VnV Plan.)

One way that the VnV plan was different from the activities conducted was that we initially planned to test all our modules, prior to us determining and forming our modules. This caused an issue when trying to actually conduct the VnV, as the modules were designed without having the testing in mind. This led us to have modules that were difficult to directly test and have concrete results for. As a result, one modification to the plan and overall system, is that certain modules had to be refactored as they could not be effectively tested and did not have any code we can point to. Furthermore, our VnV plan was dependent on a specific solution, that was vastly modified throughout the year. The VnV plan should have been written in a more generalized manner such that it would be flexible for any solution. For example, earlier in the year our team thought that a feedback module would be an appropriate way to train our AI model, however, through more research and various experiments, we determined that hypertuning would lead to better results. We believe we could not have anticipated this, as based on our AI knowledge at the time, we believed this would lead to the best design and implementation. Looking back, it's important for us to make a testing plan that depends more on the results and functionality that our system will exhibit rather than how it will be implemented as this often changes.

Fiza's Reflection

1. What went well while writing this deliverable?

   One aspect that went well while writing this deliverable was the clear structure provided by the VnV template and the previous planning documents. Since the system architecture, modules, and test plans were already defined in earlier deliverables such as the Module Guide and VnV Plan, it was easier to connect the implemented tests to the system components. Collaboration within the team also worked well, as each member was responsible for specific modules and testing sections, which allowed the work to be divided efficiently and integrated into a consistent report format.

2. What pain points did you experience during this deliverable, and how did you resolve them?

   One challenge during this deliverable was ensuring consistency between the different sections of the report, such as aligning manual tests, system tests, and module traceability with the modules defined in the Module Guide. Another difficulty was formatting complex LaTeX tables and ensuring they

compiled correctly while maintaining a consistent style across the document. These issues were resolved through iterative debugging of the LaTeX code and by coordinating with teammates to ensure the testing sections aligned with the previously defined modules and test cases.

Moly Mikhail's Reflection

1. What went well while writing this deliverable?

   I think many aspects went well while writing this deliverable. Writing the unit tests for the backend was a smooth process as I was highly familiar with the code base and the different branches to be covered. Another aspect that went well was documenting our unit test results, this also provided another opportunity for me to review the unit tests so far and to determine if any tests were missing and add them.

2. What pain points did you experience during this deliverable, and how did you resolve them?

   One pain point I experienced writing this deliverable is determining the difference between the unit test section and automated testing. At first I was confused on the difference; however, after discussing with our TA, it was clearly they are highly related. Furthermore, it was also difficult determining which aspects of the system should be uni tested, verses manual tested. This was also cleared up through discussions with Chris. We decided the for integration testing it would be more effective to do it manually rather than to implement Selenium.

Dhruv Sardana's Reflection

1. What went well while writing this deliverable? I was responsible for writing the frontend modules tests. It helped me understand how to set up tests in Jest and React Testing Library and how to run coverage. Additionally, it helped me identify gaps in the application that were uncovered through testing.

2. What pain points did you experience during this deliverable, and how did you resolve them?

   The challenges I faced were mainly related to setting up the testing environment and writing tests for React components that involve asynchronous operations, such as mocking API calls. I resolved these issues by referring to documentation and examples for Jest and React Testing Library, to ensure

that the tests were correctly structured and covered the necessary functionality. Also another major challenge was to understand how to split up unit tests by modules

Nawaal Fatima's Reflection

1. What went well while writing this deliverable? The team was better prepared to work together. We split the doc as per usual and got to work, updating each other as often as possible. Our communication was stronger with each other this time around.

2. What pain points did you experience during this deliverable, and how did you resolve them? My VSCode decided to crash every time I tried to work with it, so I had to ask my teammates to make my edits for me which was really annoying :(The tests we wrote for the VnV Plan didn't accurately encompass the solution we ended up building. They seemed disconnected from the solution itself as some parts had vastly changed from how we thought we'd implement it. This was frustrating because we knew our product worked, we just couldn't show it based on a plan we made before we started coding.

Casey Francine Bulaclac's Reflection

1. What went well while writing this deliverable?
   One aspect that went well while writing the Verification and Validation report was being able to directly test our implemented modules and observe the outcomes of the test cases. Running the tests allowed us to clearly see which tests passed and which failed, which helped us better understand the current performance of our tool. This process was valuable because it highlighted areas where the system was functioning correctly as well as areas that could be improved for Revision 1. Another positive aspect was conducting user testing. Gathering feedback from users helped us identify what parts of the tool were working well and which parts needed improvement. Seeing how users interacted with the system provided practical insight and helped validate whether the tool behaved as expected from a user perspective.

2. What pain points did you experience during this deliverable, and how did you resolve them?
   One challenge we encountered was that some modules originally described in the design documentation were not fully implemented in the final system. Because of this, we could not create meaningful verification tests for those modules as originally planned. This required us to revisit the testing plan and modify the VnV report so that the test cases only covered modules that

were actually implemented and testable. To resolve this issue, we updated the relevant documentation to ensure consistency across all project deliverables. This involved adjusting parts of the MIS and MG documents so that they accurately reflected the system that was ultimately implemented. Although this required additional revisions, it ensured that the documentation, design descriptions, and testing procedures were all aligned with the final state of the project.